

Mental Navigation in the Primate Entorhinal Cortex

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Part 1 - Experimental Background

The Core Questions

- **Behavioral Capability** - Can monkeys navigate without visual feedback?



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- **Behavioral Capability** - Can monkeys navigate without visual feedback?
- **Neural mechanism** - What neural structures support this?
- **Role of structure** - How do these structures improve performance?



Cognitive Maps

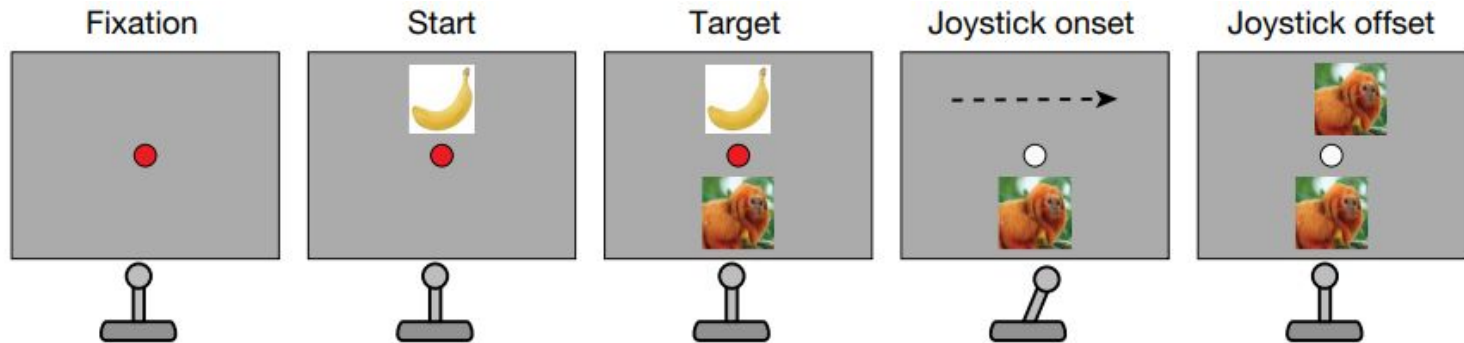
- An internal representation of relationships between states
- Encodes **structure**, not just stimulus-response pairs
- Supports inference about novel situations

Why the Entorhinal Cortex?

- Known for grid cells and path integration
- Encodes spatial and abstract variables
- Candidate substrate for cognitive maps

Experimental Setup

- A monkey is seated in front of a screen
- Given a joystick to control the movement of a dot along a line segment
- Visual landmarks are displayed on the screen
- Joystick speed is fixed, so distance is encoded by movement time
- **Objective** - From a *starting landmark*, use the joystick to navigate to a *target landmark*.

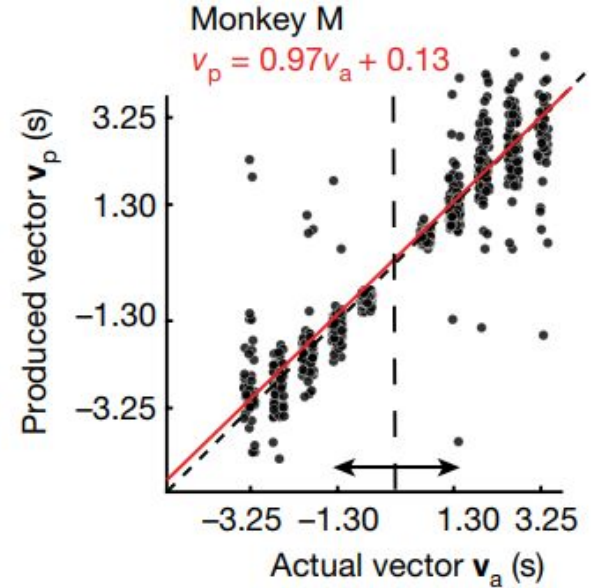
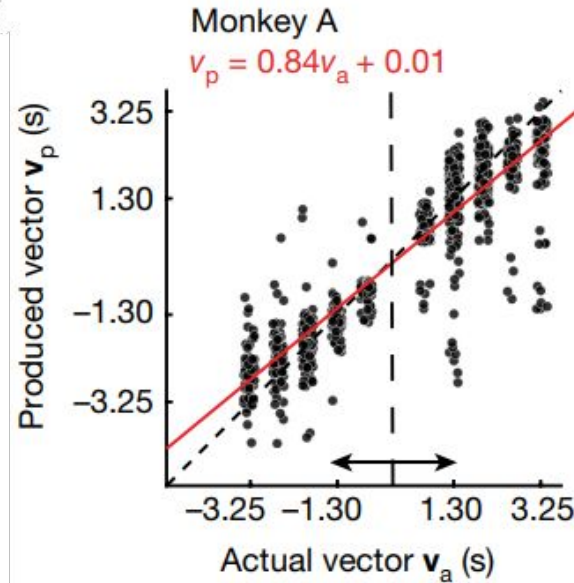


Learning vs Mental Navigation

- **Learning Phase** - Landmarks are visible during movement
 - The monkey learns landmark spacing and joystick dynamics
- **Mental Navigation (MNAV) Phase** - Landmarks are not visible during movement
 - The monkey must navigate without visual feedback.

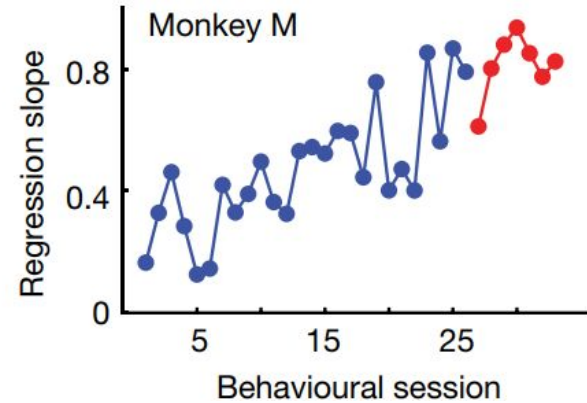
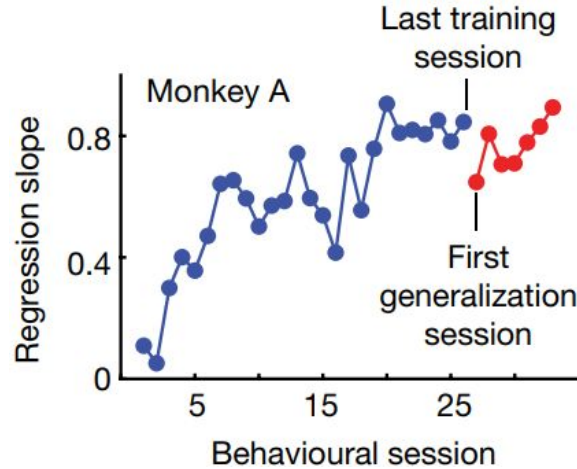
Behavioral Performance

- Accurate production of distances
- Linear relationship: produced vs actual
- High performance across sessions



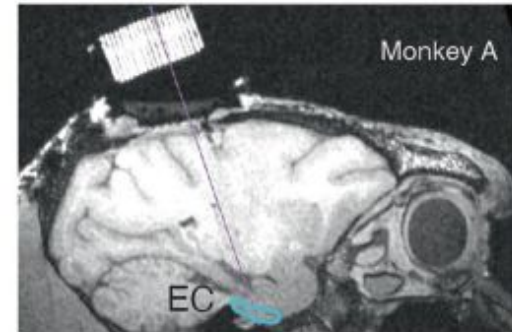
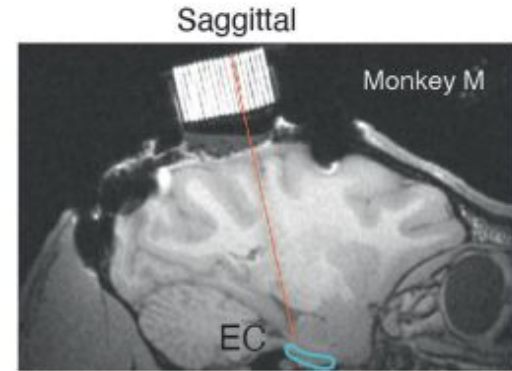
Generalization

- Only half of the landmark pairs are trained
- The remaining pairs are held out during training
- The monkeys immediately generalize to new pairs during MNAV



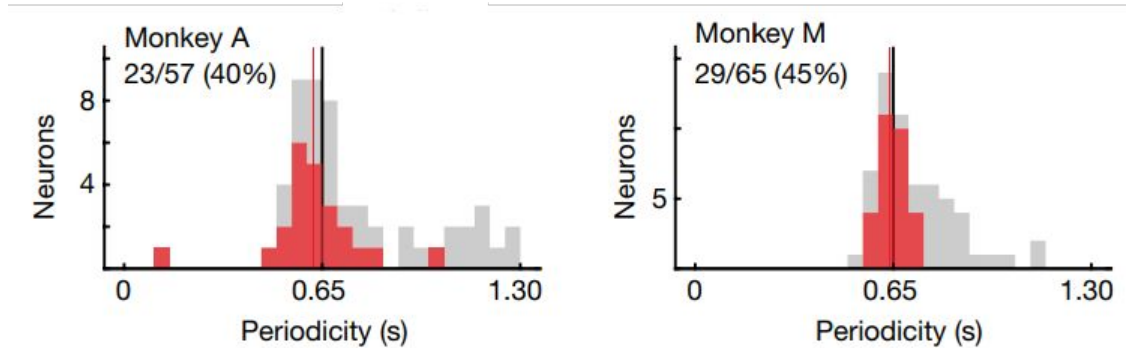
Neural Recordings

- Unit recordings in the entorhinal cortex
- Analysis focused on the navigation epoch
- No visual input during this period



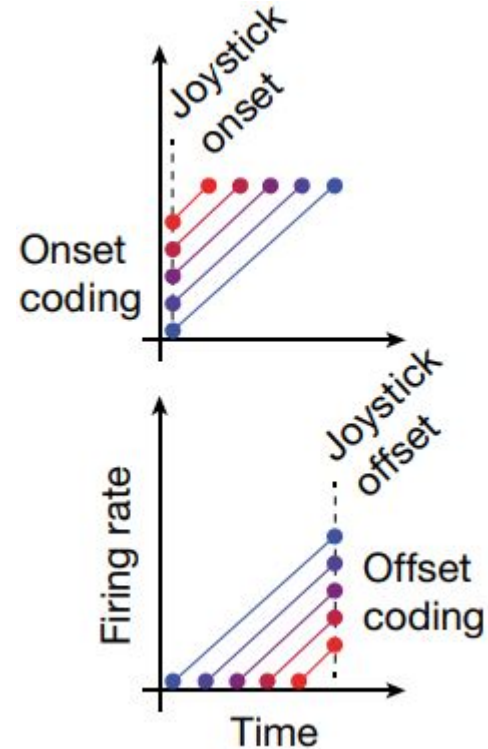
Neural Signature 1 - Periodicity

- Transient firing *bumps*
- Period matches landmark spacing
- Generated without sensory cues



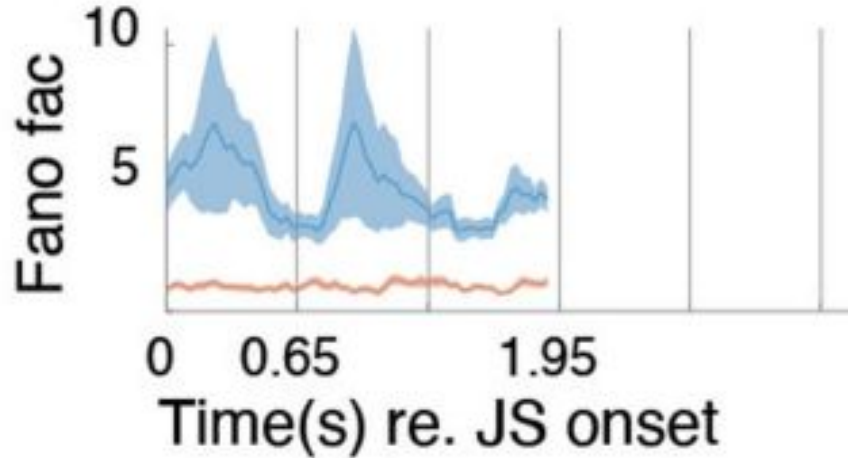
Neural Signature 2 - Ramping

- Firing rate ramps with distance
- Strong encoding near joystick offset
- Distinct from classical timing ramps



Neural Signature 3 - Variability

- Trial-to-trial variability is not uniform
- Variability is reduced near landmarks
- This predicts improved behavioral accuracy



Experimental Summary

- Internal representation is structured
- Neural activity with periodic and ramping dynamics coexist
- Landmark structure reduces variability

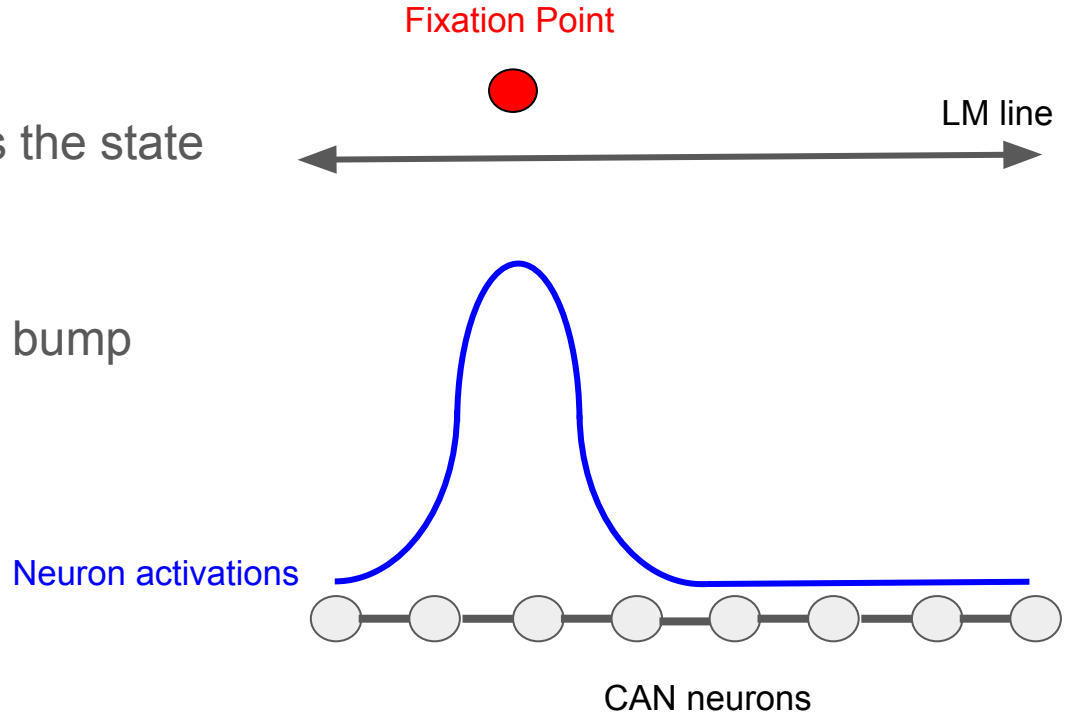
Part 2 - Computational Model

Modeling Goal

- Explain neural dynamics mechanistically
- Account for reduced variability
- Link neural activity to behavior

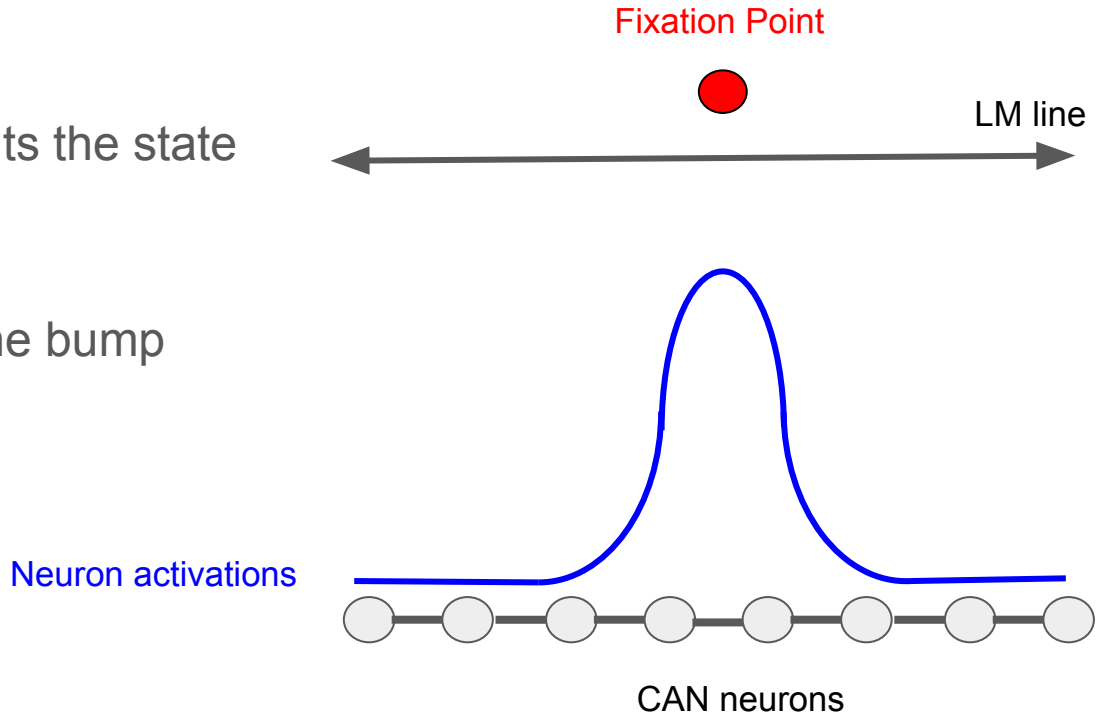
Continuous Attractor Networks (CANs)

- Stable low-dimensional manifold
- The activity bump represents the state
- The velocity input moves the bump



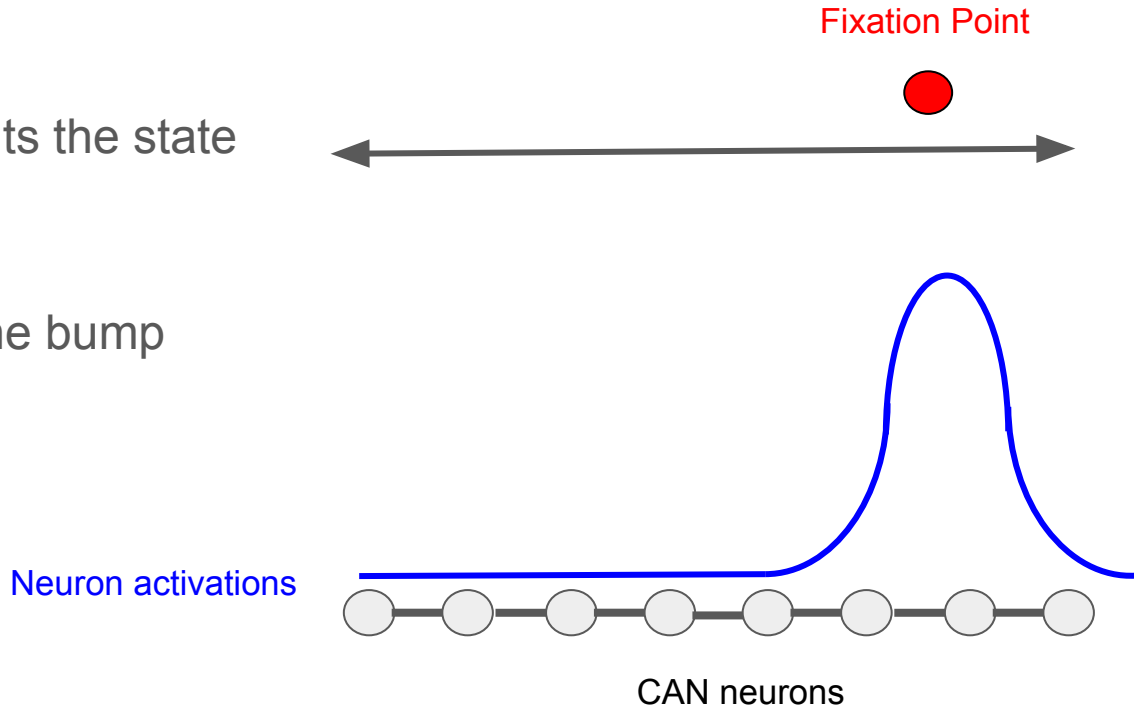
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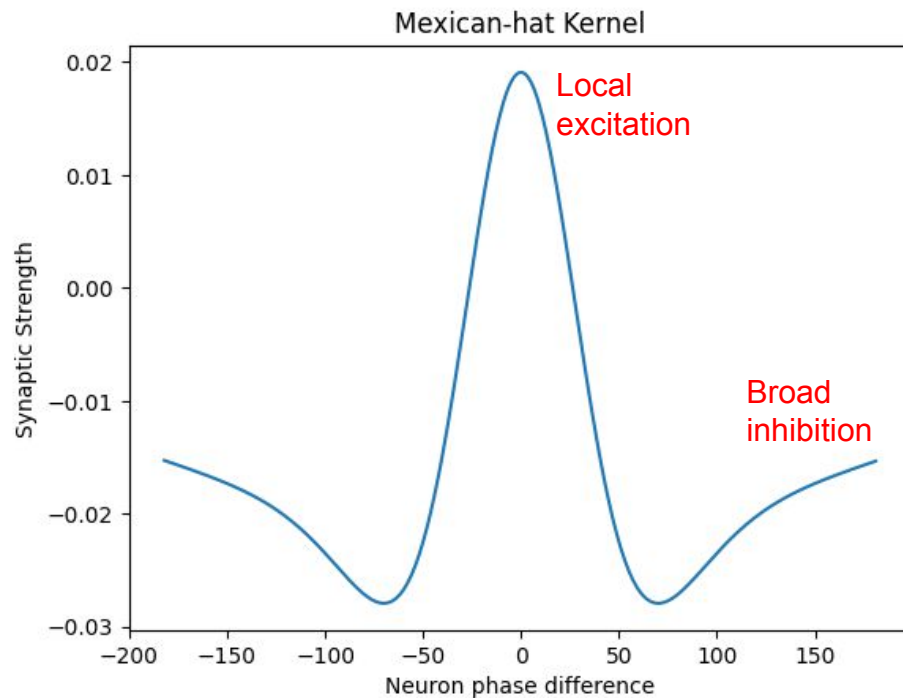
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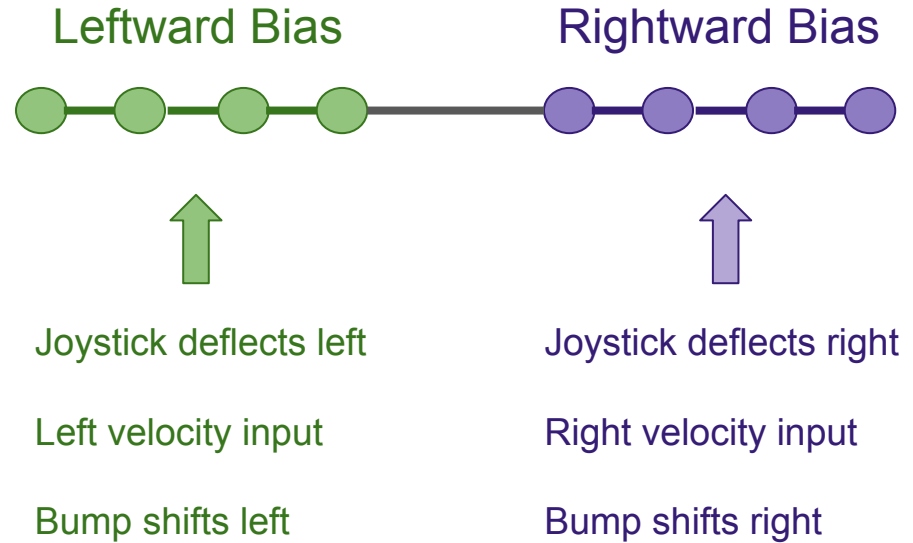
CAN Architecture

- 1D ring topology
- “Mexican-hat” recurrent connectivity



CAN Architecture

- 1D ring topology
- “Mexican-hat” recurrent connectivity
- Left / right populations biased by velocity



State Representation

- Network state = bump phase
- Phase tracks elapsed distance
- Noise causes phase diffusion

Gaussian noise is added to the velocity input at each timestep

```
# Construct velocity vector  
v = wlm_speed * (1.0 + wm * np.random.randn(N_t))
```

True speed

Scaling param

Gaussian noise

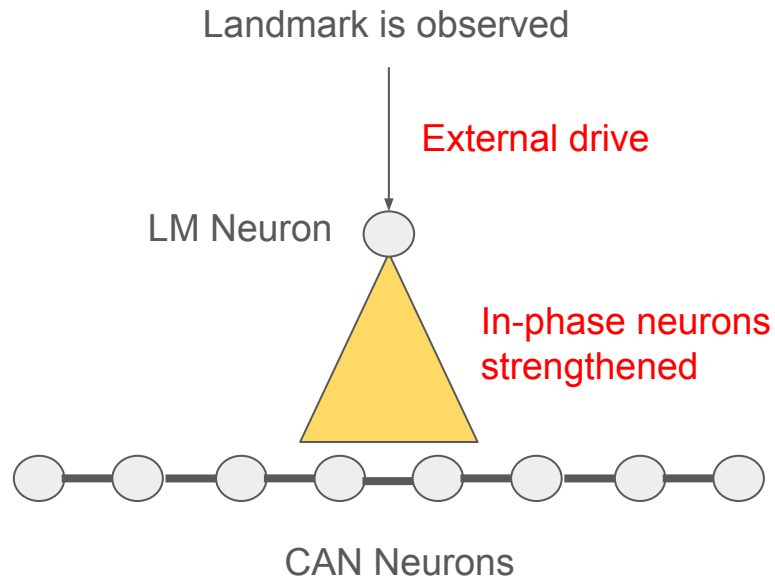
CAN Without Landmarks

- Smooth bump drift
- Noise accumulates continuously
- Variability grows linearly with distance

$$\sigma_{\text{No LM}} \propto t$$

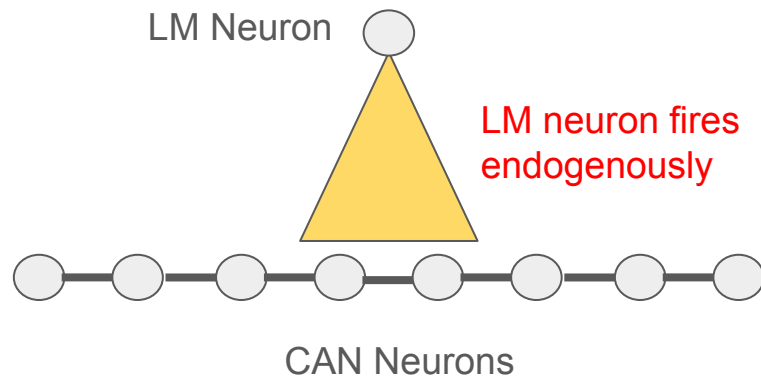
Landmark Neurons

- Neurons that fire at specific phases
- Driven externally during learning



Landmark Neurons

- Neurons that fire at specific phases
- Driven externally during learning
- Driven internally during recall



Learning the Landmark Structure

- Hebbian / Oja learning rule
- Strengthen phase-matched inputs
- External landmarks → Internal landmarks

Oja rule update

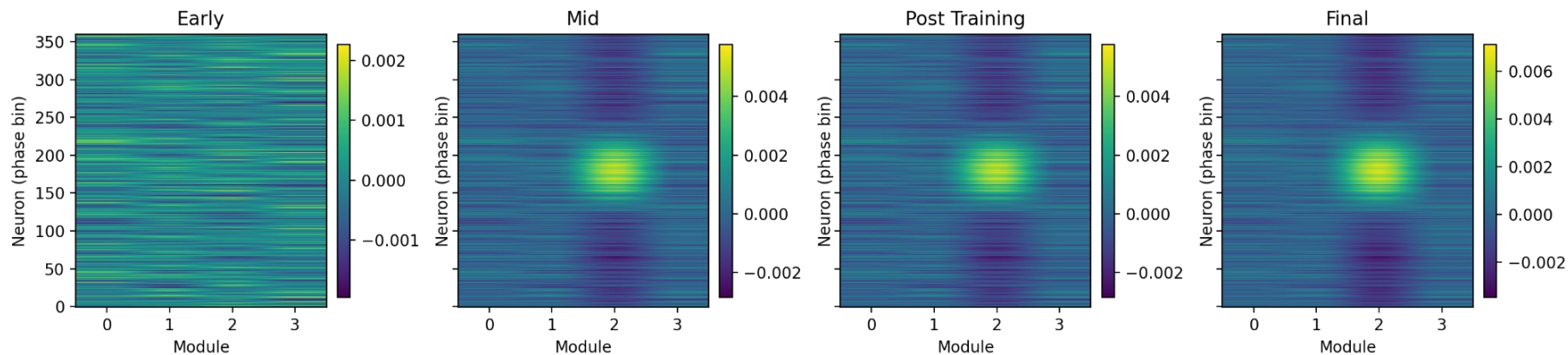
```
weights += LR * lm_input * (module_activations - lm_input * weights)
```

Synapse weights

LM neuron
activation

CAN neuron
activations

Landmark Neuron - CAN Synaptic Weights



Reproduction of *Fig. 4b* in the paper

- Synapses select the grid cell module with the matching **periodicity**
- Synapses select the neurons with the matching **phase**

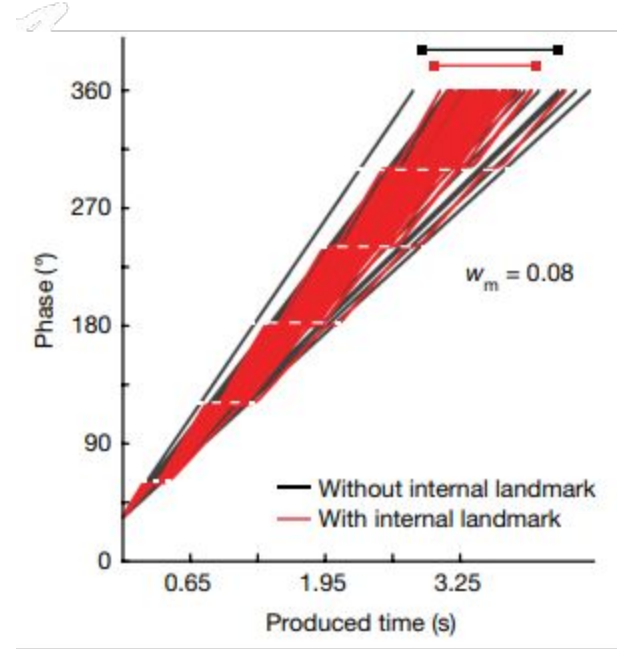
CAN With Internal Landmarks

- Landmark input stabilizes the bump
- The bump's motion slows near landmarks
- Produces reset-like dynamics

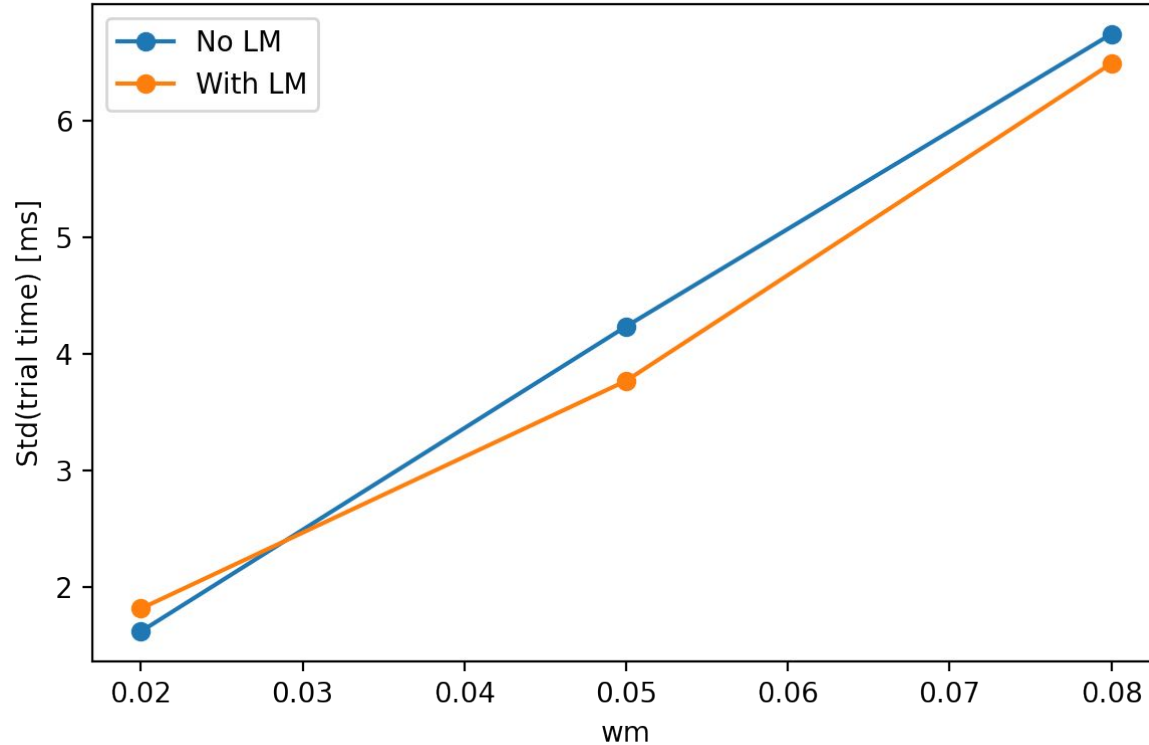
$$\sigma_{\text{LM}} \propto \sqrt{t}$$

Variability Scaling

- No landmarks - Error grows linearly
- With landmarks - Error grows sublinearly
- With landmarks behavior matches the monkey's



Variation in Duration as a function of Input Noise



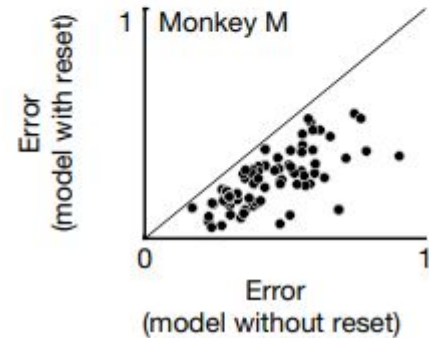
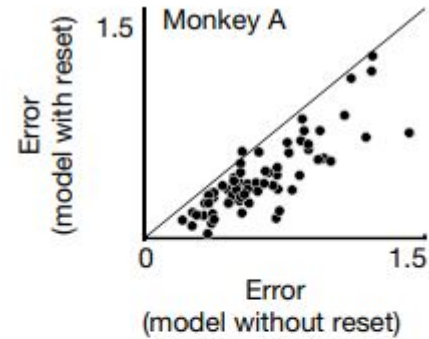
The CAN with LM neurons has lower variability as noise is increased relative to the CAN without LM neurons

Periodic Variability

- Variability oscillates in time
- Period of the oscillations equals the landmark spacing
- This is predicted by the model and seen in neurons

Linking the Model to Behavior

- Bayesian observer models
- Landmark-reset model fits better
- Mechanistic explanation of performance



What the Model Explains

- Periodic neural activity
- Ramping distance signals
- Reduced behavioral variability

Broader Implications

- Mental counting and rehearsal
- Abstract cognitive navigation
- General principle for structured computation

Summary

- Memory actively shapes dynamics
- Structure reduces uncertainty
- Cognitive maps are computational tools